

Two universal vertices and a false spectral classification

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1 The original conjecture

For a simple graph G , let $q(G)$ be the largest eigenvalue of its signless Laplacian $D(G) + A(G)$. A forbidden subgraph need not be induced. The graph $K_{s,t}^+$ is obtained by adding an edge in the part of size s of $K_{s,t}$.

Conjecture 5.1 of Zheng, Li and Li [1]. For $2 \leq s \leq t$ and $n \geq s + t$, every n -vertex $K_{s,t}^+$ -free graph of maximum q is either the join of K_{s-1} with a nearly $(t-1)$ -regular triangle-free graph, or the join of I_{t-1} with a nearly $(t-1)$ -regular triangle-free graph.

Here a join adds all edges between its two parts. The residual graph has the order needed to make n vertices. “Nearly regular” means regular except that one vertex has degree $t-2$ when the parity of its order makes $(t-1)$ -regularity impossible. For $s = t = 3$, this exception does not arise: the proposed graphs are

$$K_2 \vee R \quad \text{or} \quad I_2 \vee R, \quad R \text{ triangle-free and 2-regular.}$$

The conjecture is false. We first prove its original six-vertex counterexample, including uniqueness, without a computer search. We then explain the structural reason the proposed families fail: with two universal vertices, triangles in the remainder are allowed, while four-cycles are forbidden. This gives a complete sharp theorem within that class and an obstruction at arbitrarily large orders. The earlier six-vertex result is available in the public case package.

2 A short proof of the original counterexample

Theorem 1. *The unique six-vertex $K_{3,3}^+$ -free graph of maximum q is*

$$H = K_2 \vee (K_3 \sqcup K_1), \quad q(H) = 5 + \sqrt{13}.$$

It belongs to neither family in the conjecture.

Proof. The vertex coming from K_1 has degree two. Any $K_{3,3}$ on these six vertices would give every vertex at least three neighbors, so H is even $K_{3,3}$ -free. On vectors constant on its two universal vertices, its triangle, and its remaining vertex, $D + A$ acts by

$$\begin{pmatrix} 6 & 3 & 1 \\ 2 & 6 & 0 \\ 2 & 0 & 2 \end{pmatrix}.$$

Its characteristic polynomial is $(x-4)(x^2-10x+12)$. The positive Perron vector lifts to the whole graph, giving $q(H) = 5 + \sqrt{13} > 17/2$.

Now let G be any feasible graph on six vertices. If some vertex has degree at most two, completing the other five vertices to a clique and its own neighborhood to two vertices embeds

G in a copy of H . Adding a missing edge strictly increases the spectral radius up to this connected supergraph. Thus $q(G) \leq q(H)$, with equality only when G is itself H .

It remains to treat minimum degree at least three. The complement then has maximum degree at most two. If one of its components has at least four vertices, it contains a four-vertex path. Consequently G is a subgraph of $K_2 \vee P_4$, since P_4 is self-complementary. Give the two universal vertices weight four, the two ends of P_4 weight two, and its two internal vertices weight three. Applying $D + A$ gives weights 34, 17, 25 on these three types, respectively. These are at most $17/2$ times the assigned weights, with a strict inequality on the last type. Positivity and connectedness therefore give $q(K_2 \vee P_4) < 17/2$.

Otherwise all complement components have order at most three. Their vertices can be divided into two unions of components of order three unless the complement is three disjoint edges. Such a division gives a spanning $K_{3,3}$ in G . Any additional edge within either part would give the forbidden graph, so feasibility forces $G = K_{3,3}$, with $q = 6$. In the sole exceptional case G is the complement of three disjoint edges, a 4-regular graph with $q = 8$. Neither can compete with H .

At this order the conjectured residual graph must be C_4 . The candidate $K_2 \vee C_4$ is infeasible: a universal vertex and two opposite cycle vertices have three common neighbors and an edge within their triple. The other candidate, $I_2 \vee C_4$, is 4-regular and has $q = 8$. This also directly refutes the conjecture. $\square$

The spectral comparison just used is elementary. If M is a nonnegative matrix and w is positive, then $Mw \leq cw$ bounds its spectral radius by c . When M is irreducible and one coordinate is strict, multiplying by a positive left Perron vector makes the bound strict. The same argument gives strict monotonicity under adding an edge to a connected supergraph.

3 The natural class: two universal vertices

A vertex is universal if it is adjacent to every other vertex. Thus graphs with two universal vertices are exactly the joins $K_2 \vee R$. In this whole class there is a complete answer.

Theorem 2. *The graph $K_2 \vee R$ is $K_{3,3}^+$ -free if and only if R has maximum degree at most two and has no C_4 component. Every such graph is also $K_{3,3}$ -free.*

For each $n \geq 7$, the maximum of q among these n -vertex graphs is

$$\frac{n + 6 + \sqrt{n^2 - 4n + 20}}{2}.$$

The complete equality class consists of the joins of K_2 with disjoint unions of cycles, none of length four, on $n - 2$ vertices.

Proof. A graph contains $K_{3,3}^+$ precisely when a triple containing an edge has three common neighbors. Common neighbors lie outside the triple because the graph has no loops.

Write u, v for the universal vertices. If a vertex of R has three neighbors in R , it and u, v violate this criterion. If R has a four-cycle, a universal vertex and two opposite cycle vertices violate it. These prove necessity.

Conversely, suppose R has maximum degree two and no four-cycle component. A triple containing both universal vertices has at most two common neighbors. A triple containing exactly one universal vertex can have the other as a common neighbor, and at most one more in R : two common neighbors in R would make a four-cycle, necessarily an entire component under the degree bound. A triple entirely in R has no common neighbor there, again by the degree bound. Thus every triple has at most two common neighbors. This proves even $K_{3,3}$ -freeness.

For the spectral assertion, let q_* be the displayed number. Assign weight one to every vertex of R and weight $(q_* - 6)/2 > 0$ to each universal vertex. If every residual degree is two, applying $D + A$ multiplies these weights by q_* : the two required equations are

$$q_*h = nh + n - 2, \quad q_* = 2h + 6,$$

whose larger solution is exactly the displayed root. A smaller residual degree strictly lowers its own coordinate. The positive vector comparison therefore proves $q \leq q_*$, with equality if and only if every residual degree is two.

A finite 2-regular graph is a disjoint union of cycles; the feasibility criterion removes exactly the four-cycles. Such a remainder always exists for $n \geq 7$, by taking C_{n-2} . This proves both attainment and the complete equality statement. $\square$

This is a sharp classification on the full two-universal-vertex class, not a collection of examples within it. The six-vertex exception is already covered by Theorem 1; there is no feasible 2-regular remainder of order four. At order five, the same formula and equality rule give K_5 .

4 Why no order threshold repairs the original families

For $n = 8$ and for every $n \geq 10$, Theorem 2 permits

$$K_2 \vee (C_3 \sqcup C_{n-5}).$$

Its spectral radius is q_* , but it is outside both source families. Indeed, its two universal vertices are determined by degree; deleting them leaves a triangle. This precludes the triangle-free residual family. The independent-pair family has no universal vertex at these orders.

Every graph in the clique-pair source family has the same value q_* , because its remainder is 2-regular. Every graph in the independent-pair source family has smaller value: adding the edge between that pair strictly increases q and gives the corresponding clique-pair graph.

There is now no need to know the unrestricted maximum. If it equals q_* , the displayed graph is an outside maximizer. If it is greater, all maximizers are outside the source families. In either case the original all-maximizers claim fails. Since every $n \geq 10$ is covered, increasing only the order threshold cannot repair it. The displayed construction is intentionally not used at $n = 9$, where its remainder would contain a four-cycle.

The natural remaining question is whether large-order unrestricted maximizers must have two universal vertices. The present theorem does not establish that reduction. It resolves the original conjecture negatively and completely determines the structurally motivated class, while leaving a corrected all-graph classification open.

References

- [1] J. Zheng, Y. Li and H. Li. *The signless Laplacian spectral Turán problems for color-critical graphs*. arXiv:2504.07852v1, Section 5.1, Conjecture 5.1.